

COMP30850 Assignment - Spring 2025/26

Deadline: Sunday 8th March 2026

Overview

Analysing bike rental data can offer valuable insight into how people move around urban spaces, revealing patterns of everyday travel. This assignment involves applying network analysis and visualisation techniques to a Dublin bike rental dataset to identify hub stations, frequent routes between bike stations, and temporal trends.

The assignment should be implemented as a single Jupyter Notebook (not a script). Your notebook should be clearly documented, using comments and Markdown cells to explain the code and interpret the results of your analysis.

The data for the assignment is available at the link below.

<http://mlg.ucd.ie/modules/COMP30850/bike-data.zip>

This data contains two separate files:

1. *stations.csv*: Information about individual bike rental stations, one station per row with a unique identifier.
2. *rentals.csv*: List of bike rentals, one row per trip, including the identifiers of the start and end stations.

Tasks

Complete the following four tasks:

1. Static Network Construction

- a) To analyse the station-to-station movement, create an appropriate network representation from the raw data with the following properties:
 - Each node represents a unique station, identified by a unique identifier. It should also have attributes containing the station's name and area.
 - Each edge indicates one or more trips from an origin station to a different destination station.

2. Static Network Characterisation

- a) Apply a range of different methods to characterise the structure and connectivity of the bike-trip network created in Task 1.
- b) Apply several different centrality measures to identify hub stations in the bike-trip network. Discuss how the centrality scores might be interpreted in each case.
- c) Identify any stations with significant positive net flow (more arrivals than departures) and significant negative net flow (more departures than arrivals).

3. Static Network Visualisation

- a) Extract a subgraph from the bike-trip network created in Task 1 containing only the stations located in the 'Northside' area of Dublin. Export this subgraph as a GEXF file.
- b) Load the GEXF file in Gephi. Use Gephi's layout functionality to create several useful visualisations of the subgraph. Your visualisations should highlight different aspects such as the network's overall connectivity structure. Export the final visualisations as PNG files.

4. Dynamic Network Analysis

- a) Using the dates associated with each trip, construct a set of daily time networks representing station-to-station movement on each day.
- b) Apply various methods to characterise the global structure and connectivity of the daily networks. Compare these network-level metrics across different days and discuss any temporal patterns you observe.
- c) By analysing station-level activity over time, identify stations where capacity should be increased by the bike rental operator.

Use of Generative AI:

- You may use generative AI tools for this assignment (e.g., ChatGPT, Copilot, Gemini), provided you disclose their use as described below.
- At the beginning of your notebook, include an "AI Use Statement" in a Markdown cell. This is required even if you do not use any generative AI tools. If you do use generative AI, list the tool (and model, if known) and describe how it was used (e.g., understanding data, debugging or generating code). Penalties will apply if this statement is missing or inaccurate.

Guidelines

- The assignment should be completed **individually**. All submissions will be subject to plagiarism checking. Any evidence of plagiarism can result in a 0 grade.
- The grade awarded will be based on the complexity and appropriateness of the analysis and the level of detail.
- See the associated Grading Rubric document for a detailed breakdown of marks.
- Submit your assignment via Brightspace. Your submission should be in the form of a single ZIP file containing your notebook file (.IPYNB), the GEXF file exported in Task 3, and the PNG files for the visualisations created in Gephi.
- Penalties will apply for late submissions after the specified deadline:
 - 1-5 calendar days late: 1 grade point deduction, e.g. B to B-
 - 6-10 calendar days late: 2 grade point deduction, e.g. B to C+
 - Assignments will not be accepted any later than 10 calendar days without Extenuating Circumstances formally approved by UCD.